

Poster: Real-time Material and Texture Recognition Using Visible Light Communication

Jiarong Li^{1,2,†}, Chenxin Liang^{1,†}, Zixuan Xie¹, Xiaojun Liang^{2,*}, Wenbo Ding^{1,2,3,*}, Jian Song^{1,2,5},
Xiao-Ping Zhang^{1,3,4}

¹Tsinghua-Berkeley Shenzhen Institute, Shenzhen International Graduate School, Tsinghua University, ²Peng Cheng Laboratory, ³RIOS Lab, ⁴Ryerson University, ⁵Department of Electrical Engineering, Tsinghua University
{li-jr22, liangcx23, xie-zx21}@mails.tsinghua.edu.cn, liangxj@pcl.ac.cn
ding.wenbo@sz.tsinghua.edu.cn, jsong@tsinghua.edu.cn, xzhang@ee.ryerson.ca

ABSTRACT

In response to the challenges presented by conventional material and texture recognition methods, our research introduces a system using visible light communication (VLC) technology. This approach provides a non-contact, non-destructive, dual-functional solution, overcoming the limitations of cost, safety, and environmental adaptability associated with traditional methods. Through a comprehensive design integrating hardware and software, our system utilizes VLC for precise and efficient recognition. Extensive testing confirms its effectiveness, achieving 97.7% accuracy in material identification and 93.8% in texture detection. This study highlights VLC's potential in enhancing automated recognition systems across various applications.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing.

KEYWORDS

Visible Light Communication, Material Classification, Texture Recognition, Machine Learning

ACM Reference Format:

Jiarong Li^{1,2,†}, Chenxin Liang^{1,†}, Zixuan Xie¹, Xiaojun Liang^{2,*}, Wenbo Ding^{1,2,3,*}, Jian Song^{1,2,5}, Xiao-Ping Zhang^{1,3,4}. 2024. Poster: Real-time Material and Texture Recognition Using Visible Light Communication. In *The 22nd Annual International Conference on Mobile Systems, Applications and Services (MOBISYS '24)*, June 3–7, 2024, Minato-ku, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3643832.3661427>

1 INTRODUCTION

In the realm of automation and intelligent devices, the capacity for environmental interaction and material perception has become increasingly pivotal [4]. This necessity spans diverse sectors, including robotics for handling and assembly, industrial production, safety

[†]Both authors contributed equally to this research. * Corresponding Author.

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MOBISYS '24, June 3–7, 2024, Minato-ku, Tokyo, Japan

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ACM ISBN 979-8-4007-0581-6/24/06...\$15.00

<https://doi.org/10.1145/3643832.3661427>

protocols, and the evaluation of food and liquid quality, underlining the versatility of non-intrusive, contactless sensing technologies [5]. These applications rely on material and texture recognition to refine operational precision and reliability, from automating production lines to ensuring product integrity.

Traditional methods for identifying materials and textures, including X-ray [2] and ultrasonic imaging [7], grapple with challenges such as high costs, health risks, and environmental sensitivity, often yielding unreliable outcomes. Similarly, technologies based on radio frequency [6], and Wi-Fi [1] encounter accuracy issues in dense or complex settings impacted by interference and spatial variances. These constraints underscore the necessity for efficient and adaptable solutions, prompting a shift towards emerging technologies like visible light communication (VLC) [3], which promises to surmount these limitations by offering a safer, more efficient recognition methodology.

Addressing these challenges, our research introduces a VLC-based system for discerning materials and textures, leveraging the potential of next-generation wireless technologies. By capitalizing on VLC's non-invasive and cost-effective attributes, the system proposes a holistic solution encompassing a VLC device, a data processing chipset, and a user interface. This design minimizes hardware dependence and operational expenses, facilitating seamless integration into existing infrastructures and achieving notable accuracy in material (97.7%) and texture recognition (93.8%). Our findings not only validate the system's efficiency and adaptability but also highlight VLC's role in advancing automated recognition technologies.

2 SYSTEM DESIGN

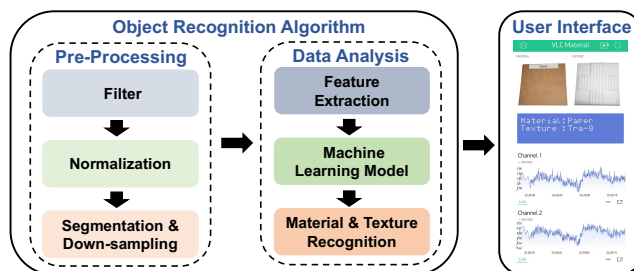


Figure 1: Object recognition workflow.

Our VLC-based recognition system combines hardware and software to modulate, capture, and interpret light signals for accurate

material and texture identification, as shown in Figures 1 and 2. The VLC device is a development board with a modulated LED and a photodiode optimized for diverse light conditions and material interactions. It is designed based on open-source OpenVLC and offers a flexible and efficient platform for data transmission essential for communication and precise recognition tasks. The captured signals are processed by a data processing chip built from ESP32-S2's architecture, featuring transimpedance amplifiers and digital integration capabilities, further supported by Internet of Things (IoT) functionalities for remote data handling and control.



Figure 2: (a) VLC device and (b) object recognition system.

Data preprocessing involves filtering, normalization, segmentation, and sampling to enhance signal quality and consistency across sensors, which is crucial for effective machine learning model training. This preprocessing paves the way for feature extraction, where key signal characteristics such as average value, spectral components, and signal power are analyzed to discern material properties and interactions. These processes culminate in the application of machine learning algorithms—random forest, decision tree, and support vector machine (SVM)—each chosen for its ability to handle the intricacies of VLC data, providing a stable, interpretable, and robust framework for material and texture recognition in IoT and edge computing scenarios. A user-friendly interface on the Blynk platform enables real-time data visualization and interaction, ensuring a seamless user experience while facilitating quick and accurate data classification.

3 EXPERIMENT

In our experiment, the VLC device was placed 15cm from target surfaces within an indoor setting, employing on-off keying (OOK) and Manchester encoding for light modulation, facilitated by a BeagleBone Black (BBB) board. Material reflections were detected by integrated receivers, with voltage data captured and visualized through an NI data acquisition card and LabVIEW, showcasing real-time IoT application capabilities. This setup tested direct current (DC), sine wave, and communication signals across 10 material types, revealing communication signals' superior balance of functionality. An assessment of four light colors using photodiodes (PDs) and solar cells identified white light and PDs as optimal for their broad spectrum and high sampling rate, respectively.

Our dataset comprised ten materials (≈ 550 samples/class) and three textures (≈ 18000 samples/class), as Figure 3 shows. Machine learning models were evaluated for material and texture recognition, achieving up to 97.7% accuracy with SVM leading, particularly

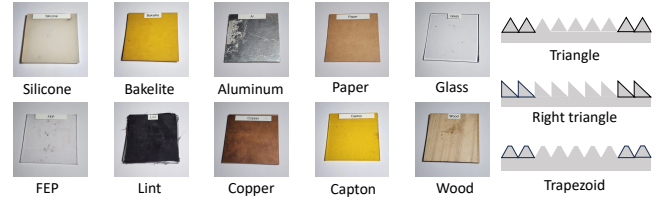


Figure 3: Material and texture specimens.

for material identification. Despite challenges in texture recognition, SVM maintained the highest accuracy at 93.8%, demonstrating the models' efficacy in handling complex data. The results highlight the system's capacity for efficient, real-time material and texture recognition with high accuracy.

4 CONCLUSION

Our study develops a VLC-based material and texture recognition system, effectively overcoming traditional methods' constraints with a non-invasive and dual-functional approach. Emphasizing software-hardware synergy, it showcases notable accuracies in identifying materials and textures. The empirical validation accentuates VLC's robust applicability in object recognition, showing a new method for automating recognition processes.

ACKNOWLEDGEMENTS

This research was supported by the Tsinghua Shenzhen International Graduate School-Shenzhen Pengrui Young Faculty Program of the Shenzhen Pengrui Foundation (No. SZPR2023008), the Shenzhen Ubiquitous Data Enabling Key Lab (No. ZDSYS20220527171406 015), the Guangdong Innovative and Entrepreneurial Research Team Program (No. 2021ZT09L197), the Major Key Project of Peng Cheng Laboratory (No. PCL2023A09), the National Natural Science Foundation of China (No. 62104125), and the Shenzhen Science and Technology Program (No. JCYJ20220530143013030).

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